IE416-Robot Programming

Course Placement: Robot Programming is an ICT and Technical Elective course for 3rd year students of BTech ICT, RAS (**R**obotics and **A**utomation **S**ystems) minor programs. It is also offered as an elective for MTech program.

Course format: It is **3 hours lecture** and **2 hour lab** every week.

Prerequisites: Python Programming, C++ Programming, and Embedded System Software

Course content: This course introduces students to the robot arm, autonomous car, and ROS (**R**obot **O**perating **S**ystem) as well as numerous available tools VSCode, Python, OpenCV, TensorFlow, and GitHub. Course focus on theory and hands-on, whereas hands-on will focus on applications and implementations motor, camera, GPS, computer vision, and artificial intelligence. Students learn how to create robot model, interface to sensors and actuators, and integrate AI/ML algorithms for autonomous robot. The course works through exercises involving a number of autonomous robots (i.e., ground) that students could eventually use in their future robotics research and innovation. The course consists of two parts which are (*i*) interactive lecture section, (*ii*) lab exercises on ROS which will provide a platform to understand a real robot from scratch.

Text Book: Yoon Seok Pyo, Han Cheol Cho, Ryu Woon Jung, and Tae Hoon Lim, Robot Programming (1st Eds.), ROBOTIS Co., Ltd. 2017, ISBN 979-11-962307-1-5 .

Reference Book:

1) Learn Robotics Programming: Build and Control AI-enabled Autonomous Robots Using the Raspberry Pi and Python by Danny Staple, (2nd Eds.), Packt Publishing, 2021
2) Robotics, Vision and Control, Fundamental Algorithms in Python, 3rd edition, Peter Corke, Springer, 2023.

Assessment method: One in-semester and a final examination, and lab assignments. Evaluation scheme: (*i*) **Second In-Sem Exam: 30%, (*ii*) End-Sem Exam: 40%, (*iii*) Labs: 30%.**

Course Outcomes: This course will give an opportunity to students to enhance their skills in basics of with a custom robotic arm and autonomous car (**PE1, PE2**). Students will gain understanding on 7 degree of freedom robotic arm, self-drive car, lane assist, cruise control, local navigating, and global navigation which are required for future automation and transportation. Students will acquire huge skills in Ubuntu, ROS, Python, OpenCV, TensorFlow, and GitHub (**PSO1, PSO2, PSO3, PE3**). Today, world leading companies such as Google, TCS, Amazon, Boston Dynamics, Softbank, DENSO, Disney, HONDA, Hyundai, etc., are looking for the people who are expertise in computer vision, artificial intelligence and robotics. This program will provide a platform to develop skills for which companies are interested to hire professionals (**PEO3**). After successful completion of the course the student will have the ability to apply engineering knowledge to identify the problem using modern tools for social development. (**PO1, PO2, PO5, PO6**). Robotics which is interdisciplinary in nature, having complexity in development required-long term goal, need to learn ethics to use in society (**PO4, PO8, PO12, PE4**). Ability to work in a group project of 3~4students in a team (**PO9, PO11, PE5**)

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PE1	PE2	PE3	PE4	PE5
X	X		X	X	X		X	X		X	X	X	X	X	X	X	X	X	X

Lecture Schedule

Sl. No	Description		No. of Lectures
1	**Robot Software**		**8**
	1.1	Robot Software Platform (ROS)	1
	1.2	Mathematics in Robotics	1
	1.3	Basics of Robotic Arm Design	1
	1.4	URDF for Custom Robotic Arm	1
	1.5	Transform between Links and Joints	1
	1.6	RVIZ Launch File and Joint Axis Test	1
	1.7	Gazebo Setup and Launching	1
	1.8	Optimization of Inertia and Collision Values	1
2	**Robot Arm Design and Programming**		**11**
	2.1	ROS Control Package and its Installation Steps	1
	2.2	Integrating Hardware Interfaces to Robot Arm	1
	2.3	Implementation of PID Controller	1
	2.4	Setting up Joint Trajectory Controller	1
	2.5	Robotics Software Tool Box	1
	2.6	Inverse and Forward Kinematics for Custom Robotic Arm	2
	2.7	Deriving DH table for Custom Robotic Arm	1
	2.8	Alexa to Actuate Robot	1
	2.9	Arduino and ROS	2
3	**Autonomous Car Design and Programming**		**8**
	3.1	Creation of Car Model	1
	3.2	Sign Board and Light Plugins	2
	3.3	Camera Sensors and Video Node	1
	3.4	Computer Vision Node	1
	3.5	Autonomous Car	2
	3.6	Own Car Model Vs Tesla	1
4	**Autonomous Car: Programming Local Navigation**		**9**
	4.1	Lane Detection with Computer Vision Techniques	1
	4.2	Custom Estimation and Data Extraction	1
	4.3	Control and Lane Assistance	1
	4.4	Detection and its Stages	1
	4.5	Localization and Classification	1
	4.6	Sign Classification Using CNN	1
	4.7	Cruise Control and Junction Navigation	1
	4.8	Traffic Light Detection Using Haar Cascades	2
5	**Autonomous Car: Programming Satellite Navigation (i.e. GPS)**		**9**
	5.1	Basics of SAT-NAV	1
	5.2	Localization	1
	5.3	Mapping	1
	5.4	Path-Planning	2
	5.5	Motion Planning	2
	5.6	Web-Based Robot Navigation	2